

Bayesian methods for the evaluation and limitation of disclosure risk in official statistics

Violeta Calian, Statistics Iceland



UNECE EXPERT MEETING ON STATISTICAL DATA CONFIDENTIALITY

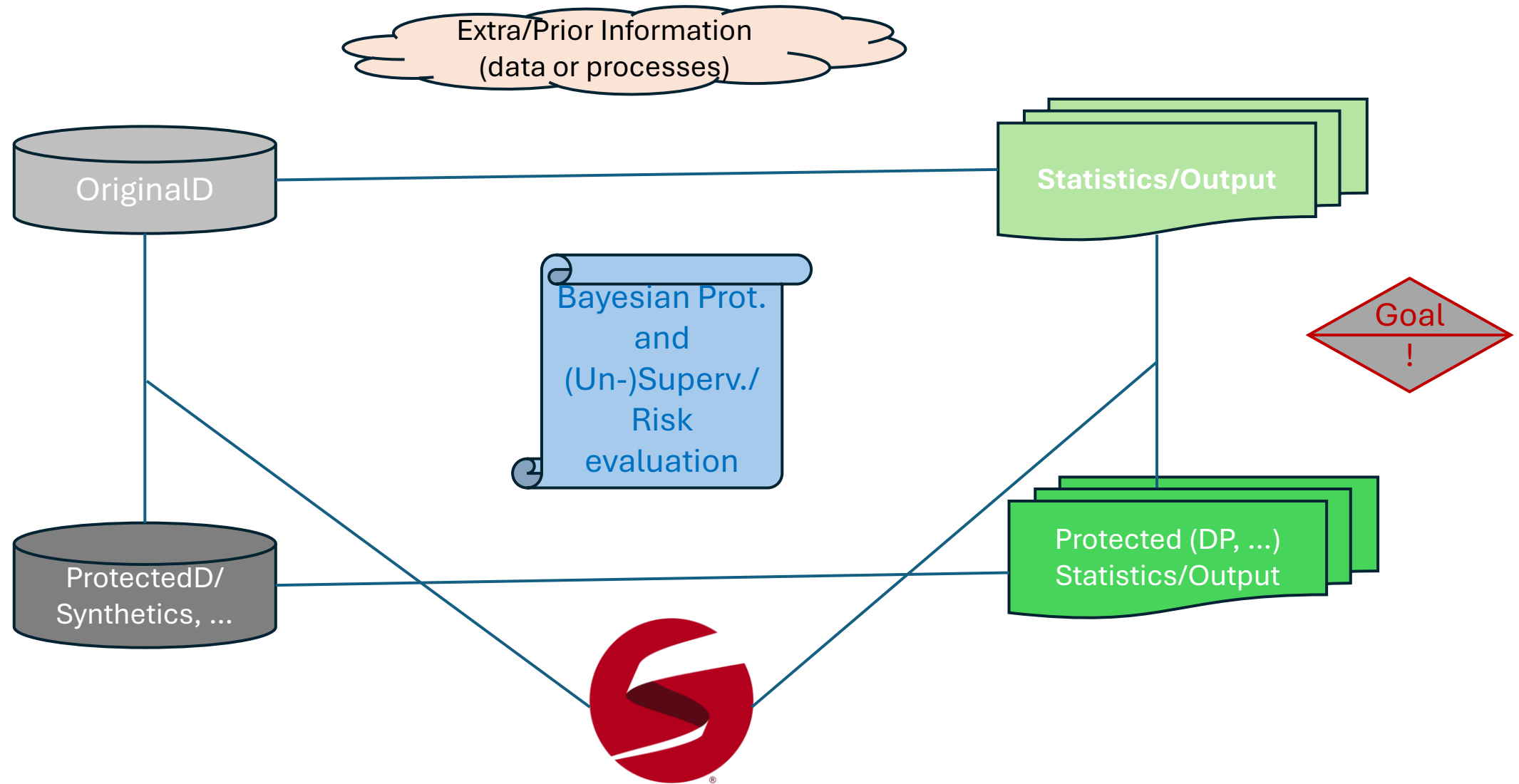
15 to 17 October 2025

Auditorium of the Poblenou Campus of Universitat Pompeu Fabra, Roc Boronat 138, 08018 Barcelona

Official Statistics: specific features of SDC

- Demands of internal experts and external researchers
- Requirements of SDC experts
- Novel theoretical and experimental developments

Bayesian role: key points



Roles of Bayes methods

- Bayes **risk** for: Original D. /Synthetic D./other protected D.
- Bayes building: synthetics & DiffPrivate output
- ML/BDeepL: for synthesizing & anomaly / risk detect & DP-operations
- Anomaly for rare patterns /unsupervised risk detection
- Validation connection!

Status

- generating multiple synthetic data sets (Graham and Penny, 2005; Graham et al., 2009; Drechsler and Reiter, 2010, 2012)
- simultaneous detection and correction of errors as well as disclosure limitation for microdata (Kim, Reiter and Karr, 2016)
- differentially private algorithms through posterior sampling (Dimitrakakis et al, 2017)
- analytical/numerical (*B*) calculation of DP parameters (*Kazan and Reiter, 2024*)

New focus and proof of principle

- the **risk** quantification and the solutions offered by Bayesian formulations for attack scenario and risk assessment
- non-parametric hierarchical Bayesian models for building data protection procedures and evaluating the disclosure risk
- Bayesian methods for *DP*
- unsupervised anomaly detection as tool for disclosure risk detection/DV

Tools:

<https://github.com/sdcTools> & also:



NEW

➤ Synthetics:

with *NPBayesImputeCat*, *synMicrodata*, *RGAN*, *synthesizer*, *synthpop*

➤ Differential Privacy:

with *Dppack*, *diffpriv*

➤ Risk and risk limits: *calculations (Bayesian style)*

Synthetics and DP (examples)

Synthetics

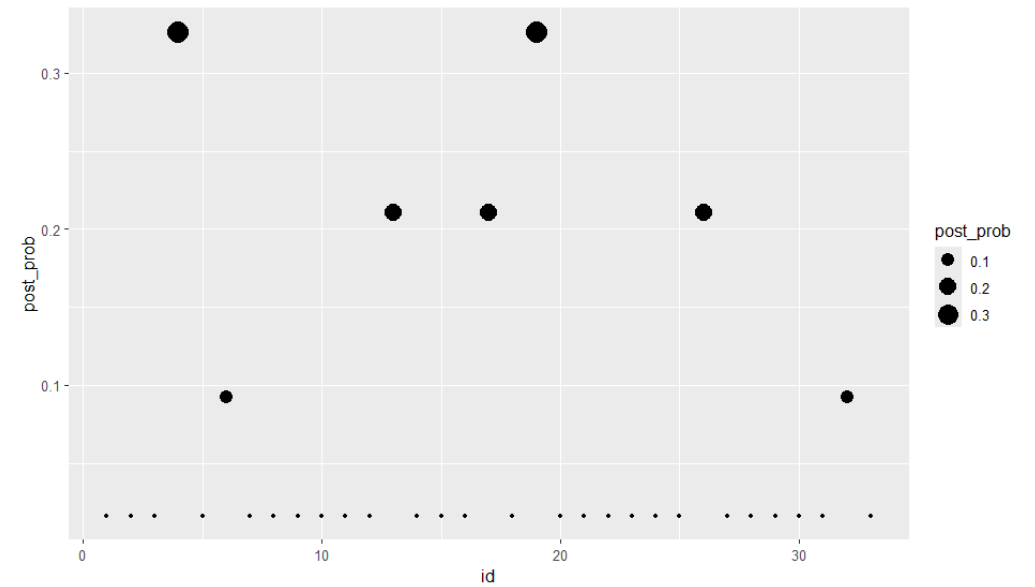
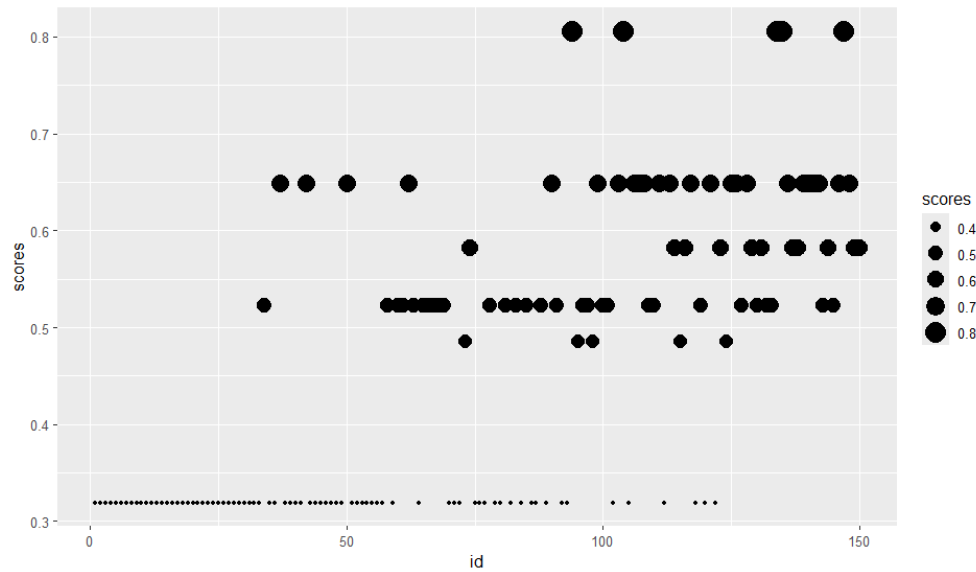
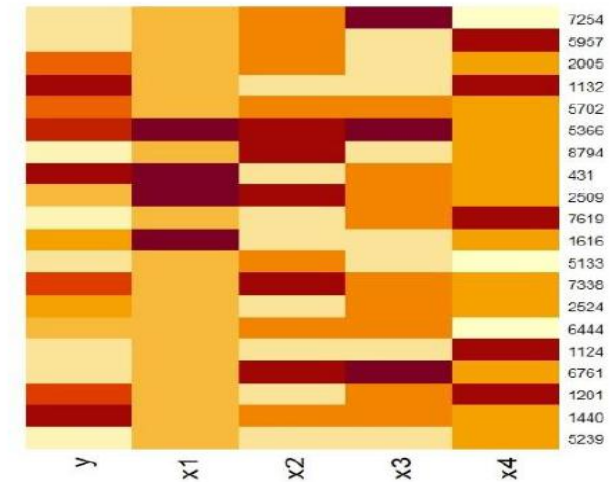
- Fig.1.

DP

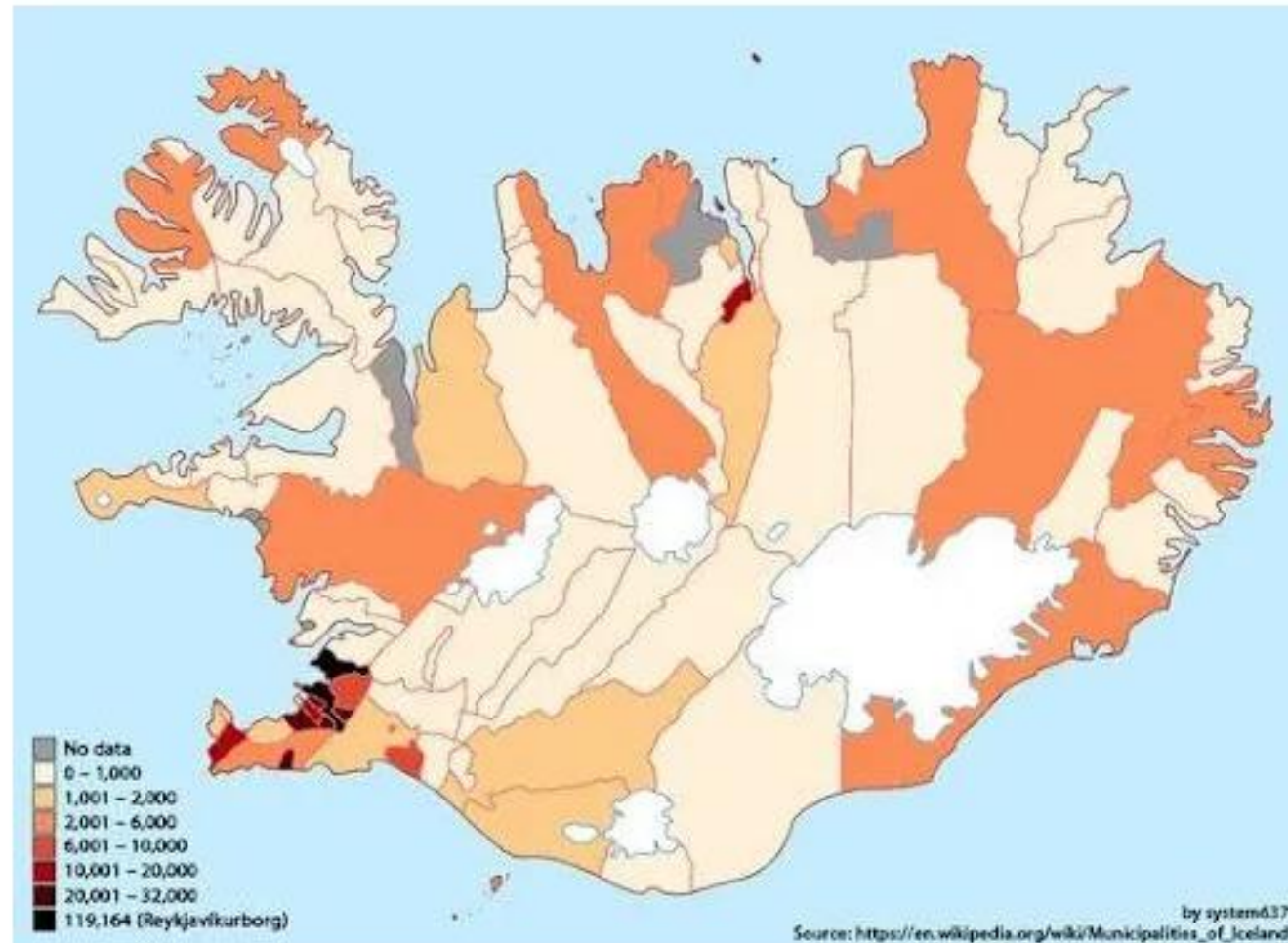
- Fig.2.

Unsupervised:

Rare patterns and anomalies



Icelandic motivation



Thank you !

- violeta.calian@hagstofa.is
- <https://github.com/violetacln/testingSDCtools>